

A Rank-Counting Reduction for Knapsack Facet Recognition with Few Candidate Coefficients

Corrected progress and an obstruction to two-anchor spanning

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Abstract

This note records a corrected analysis of a proposed fixed-parameter algorithm for recognizing whether

$$c^\top x \leq d$$

defines a facet of the binary knapsack polytope

$$P(a, b) = \text{conv}\{x \in \{0, 1\}^n : a^\top x \leq b\},$$

where the positive vector c has K distinct entries. The main conclusion is negative: the two-anchor spanning theorem used in the proposed algorithm is false, and the proposed algorithm can reject a valid facet. Thus this note does *not* solve the fixed-parameter question of Chen and Zhu.

The positive part is a replacement reduction. The affine-rank test can be expressed exactly through a Gram matrix whose entries are cardinalities of thresholded count-profile sets. These cardinalities can be obtained by a finite-difference formula from ordinary lattice-point counts in epigraph polytopes of dimension $2K$. Consequently, facet recognition for this class Turing-reduces to $O(Kn^2)$ integer-point counting queries in dimension $2K$, plus $O(n^2)$ fixed-dimensional integer-convex feasibility queries and polynomial-time rational linear algebra. If the required epigraph counting primitive is available with a uniform running time $f(K) \text{poly}(n, L)$, this gives an FPT algorithm. With only the standard fixed-dimension Barvinok-type statements, it should be regarded as partial progress rather than a full parameterized solution.

1 Status of the result

Chen and Zhu proved that knapsack-facet recognition is DP-complete in general and that, if the number K of distinct positive coefficients in the proposed inequality is fixed, the problem can be solved in time $n^{K+O(1)}$ [3]. The natural parameterized question is whether the exponent of n can be made independent of K .

This note does not settle that question unconditionally. It gives:

- (a) a counterexample to the two-anchor structural theorem in the proposed proof;
- (b) an explicit knapsack instance on which the associated anchor algorithm rejects a true facet;
- (c) a correct exact rank formula reducing the remaining difficulty to fixed-dimensional lattice-point counting.

The last item is a useful repair of the rank computation, but it is only a complete FPT solution if one supplies a uniform FPT algorithm for the counting primitive stated in Section 6.

2 Count profiles and the standard reductions

Assume, after clearing denominators, that

$$a \in \mathbb{Z}_{>0}^n, \quad b \in \mathbb{Z}_{\geq 0}, \quad c \in \mathbb{Z}_{>0}^n, \quad d \in \mathbb{Z}.$$

Variables with $a_i > b$ are fixed to zero in every knapsack-feasible point and may be deleted. After this deletion, $0, e_1, \dots, e_n \in P(a, b)$, so $P(a, b)$ is full-dimensional in the remaining variables. All statements below are relative to this reduced instance.

Let the distinct values of c be $\gamma_1, \dots, \gamma_K$, and let

$$I_j = \{i : c_i = \gamma_j\}, \quad n_j = |I_j|.$$

Inside each class I_j , order the items so that

$$a_{j1} \leq a_{j2} \leq \dots \leq a_{jn_j}.$$

Define prefix-cost functions

$$F_j(t) = \sum_{r=1}^t a_{jr} \quad (0 \leq t \leq n_j), \quad F(z) = \sum_{j=1}^K F_j(z_j).$$

The forward differences of F_j are nondecreasing, hence F_j is discrete convex. For a count vector $z \in \prod_j \{0, \dots, n_j\}$, let $p(z) \in \{0, 1\}^n$ be the classwise prefix vector

$$p(z)_{jr} = \mathbf{1}[r \leq z_j].$$

Among all subsets selecting exactly z_j items from class I_j , $p(z)$ has minimum knapsack weight $F(z)$.

Define the set of tight count profiles

$$\mathcal{Z} = \{z \in \mathbb{Z}^K : 0 \leq z_j \leq n_j, \gamma^\top z = d, F(z) \leq b\}. \quad (1)$$

Then $z \in \mathcal{Z}$ if and only if there exists a knapsack-feasible binary point x with $c^\top x = d$ and class counts z .

Proposition 2.1 (Validity reduction). *The inequality $c^\top x \leq d$ is valid for $P(a, b)$ if and only if*

$$\max\{\gamma^\top z : 0 \leq z_j \leq n_j, z \in \mathbb{Z}^K, F(z) \leq b\} \leq d.$$

If this maximum is strictly below d , the inequality is not facet-defining. The optimization problem has K integer variables and a separable convex constraint; using the epigraph formulation in Section 6, it reduces to integer linear optimization in dimension $2K$.

Proof. For fixed counts z , the prefix realization is the cheapest realization. Thus a point with candidate value $\gamma^\top z$ is knapsack feasible if and only if $F(z) \leq b$. This gives the stated maximum. If the maximum is below d , the face $\{x \in P(a, b) : c^\top x = d\}$ is empty. The epigraph formulation is given explicitly in Lemma 6.1. $\square$

Henceforth assume validity and that the maximum equals d .

3 Swap completion inside equal-coefficient classes

Canonical prefixes do not capture all tight binary points: within a class of equal candidate coefficient, a heavier selected item may be swapped for a lighter unselected item without decreasing tightness or harming feasibility.

For $r < s$ in class I_j , call the ordered pair (j, r, s) *feasible* if there exists a tight knapsack-feasible point x such that

$$x_{js} = 1, \quad x_{jr} = 0.$$

Then replacing item (j, s) by item (j, r) keeps $c^\top x = d$ and weakly decreases $a^\top x$, so both endpoints are tight. Thus the direction $e_{jr} - e_{js}$ lies in the linear span of differences of tight points. Let

$$E = \text{span}\{e_{jr} - e_{js} : (j, r, s) \text{ is feasible}\}.$$

Lemma 3.1 (Swap completion). *Let*

$$T = \{x \in \{0, 1\}^n : a^\top x \leq b, \ c^\top x = d\}$$

be the tight binary set. Then

$$\text{span}\{(1, x) : x \in T\} = \text{span}\left(\{(1, p(z)) : z \in \mathcal{Z}\} \cup \{(0, v) : v \in E\}\right). \quad (2)$$

Proof. Every vector on the right side belongs to the span of augmented tight points: $(1, p(z))$ is itself tight, and each feasible swap direction is the difference of two tight points.

Conversely, take $x \in T$ and let $z_j = \sum_{r \in I_j} x_{jr}$. If x is not the prefix vector $p(z)$, then in some class there are indices $r < s$ with $x_{jr} = 0$ and $x_{js} = 1$. Swapping s out and r in preserves $c^\top x = d$ and weakly decreases $a^\top x$, so the intermediate point remains in T . Repeating finitely many times reaches $p(z)$. Each performed swap is feasible, witnessed by the point just before the swap. Hence $x - p(z) \in E$, and therefore $(1, x)$ is in the right side of (2). $\square$

Feasible swaps can be tested without enumerating all count profiles. Fix $r < s$ in class j . For $t \in \{1, \dots, n_j - 1\}$, let

$$H_{jrs}(t) = a_{js} + \text{the sum of the } (t - 1) \text{ smallest weights in } I_j \setminus \{r, s\},$$

and set $H_{jrs}(t) = +\infty$ outside this range. On its finite domain H_{jrs} is discrete convex. Then (j, r, s) is feasible if and only if the system

$$\begin{aligned} \gamma^\top z &= d, & 0 \leq z_\ell \leq n_\ell, & \quad 1 \leq z_j \leq n_j - 1, \\ H_{jrs}(z_j) + \sum_{\ell \neq j} F_\ell(z_\ell) &\leq b \end{aligned}$$

has an integer solution. This is a fixed-dimensional separable-convex feasibility problem, again expressible by epigraph inequalities in dimension at most $2K$. There are $O(n^2)$ ordered pairs (j, r, s) .

4 The two-anchor theorem is false

The proposed proof attempted to replace all count profiles in $\mathcal{Z}$ by the minimum-cost profiles obtained after fixing two coordinates. The following example shows that no such theorem is true in the stated generality.

Let

$$K = 8, \quad n_1 = \dots = n_8 = 2, \quad \gamma = (1, 2, 3, 4, 5, 6, 7, 8), \quad d = 21.$$

Let

$$p = (1, 2, 32, 4, 8, 16, 64, 128)$$

and define univariate functions on $\{0, 1, 2\}$ by

$$\phi_j(0) = 0, \quad \phi_j(1) = p_j, \quad \phi_j(2) = p_j + 2^{8+j}.$$

Their forward differences are p_j and 2^{8+j} , so each ϕ_j is strictly discrete convex. Moreover

$$\Phi(z) = \sum_{j=1}^8 \phi_j(z_j)$$

is injective on the box: the high binary digits determine which coordinates equal 2, and the low binary digits determine which of the remaining coordinates equal 1.

Set $\tau = 227$ and

$$X_\tau = \{z \in \prod_j \{0, 1, 2\} : \gamma^\top z = 21, \Phi(z) \leq \tau\}.$$

Since any coordinate equal to 2 contributes more than 227 to Φ , the set X_τ consists exactly of binary vectors. We identify such a vector with its support. The supports in X_τ are precisely

S	$\Phi(S)$	S	$\Phi(S)$
123456	63	12468	151
12567	91	2568	154
23457	110	13458	173
13467	117	3468	180
3567	120	2478	198
		1578	201
678	208	12378	227.

Let

$$x = \mathbf{1}_{\{1,2,3,7,8\}}.$$

Then $\Phi(x) = 227$, and every other point in X_τ has smaller Φ .

Claim 4.1. *Every two-coordinate slice through x contains a point of X_τ with smaller Φ .*

Proof. The following displayed witnesses cover all pairs. For pairs contained in the support $\{1, 2, 3, 7, 8\}$ of x , use

pair	12	13	17	18	23
witness	12567	13467	1578	1578	23457
pair	27	28	37	38	78
witness	2478	2568	3567	3468	678.

For mixed pairs, with the row coordinate in $\{1, 2, 3, 7, 8\}$ and the column coordinate in $\{4, 5, 6\}$, use

	4	5	6
1	1578	1578	1578
2	12567	2478	2478
3	3567	3468	23457
7	1578	2478	2478
8	1578	3468	3468.

For pairs contained in the zero set $\{4, 5, 6\}$, use

$$45 : 678, \quad 46 : 1578, \quad 56 : 2478.$$

Each listed support has γ -sum 21, belongs to the preceding table of X_τ , agrees with x on the indicated pair, and has cost below 227. $\square$

Claim 4.2. *The point $u(x)$ in the classwise unary encoding is not in the linear span of the unary encodings of the lower-cost points in X_τ .*

Proof. All lower-cost supports in the table satisfy the additive identity

$$z_1 + 2z_2 + 3z_3 - 3z_4 - 2z_5 - z_6 + z_8 = 0. \quad (3)$$

For $x = 12378$, the left side equals $1 + 2 + 3 + 1 = 7$. The identity (3) is a linear functional of the unary encoding, obtained by assigning coefficient 1, 2, 3, -3 , -2 , -1 , 0, 1 to the level-one coordinates and coefficient zero to all level-zero and level-two coordinates. Therefore all lower unary encodings lie in a hyperplane that excludes $u(x)$. $\square$

Claims 4.1 and 4.2 contradict the proposed two-anchor spanning theorem: x is not the minimum of any two-coordinate slice through itself, but its unary row is not spanned by lower rows. Since x is the largest-cost point of X_τ , it is not contained in the two-anchor set at all, while the two-anchor set is contained in the lower-cost points.

4.1 A true facet rejected by the anchor algorithm

The preceding obstruction is not merely formal. It gives an actual knapsack facet missed by the proposed rank test.

Take eight items, all in distinct candidate-coefficient classes, with

$$\gamma = (1, 2, 3, 4, 5, 6, 7, 8), \quad p = (1, 2, 32, 4, 8, 16, 64, 128),$$

$$a_j = 1000\gamma_j + p_j = (1001, 2002, 3032, 4004, 5008, 6016, 7064, 8128), \quad b = 21227,$$

and candidate inequality

$$x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 + 6x_6 + 7x_7 + 8x_8 \leq 21. \quad (4)$$

Proposition 4.3. *The inequality (4) is valid and facet-defining for $P(a, b) = \text{conv}\{x \in \{0, 1\}^8 : a^\top x \leq b\}$.*

Proof. If $\gamma^\top x \geq 22$, then $a^\top x = 1000\gamma^\top x + p^\top x \geq 22000 > 21227$, so every feasible binary point satisfies (4). If $\gamma^\top x = 21$, then $a^\top x = 21000 + p^\top x$, and the thirteen supports listed above have p -cost at most 227, so they are tight feasible points.

The following eight tight supports are affinely independent:

$$123456, \quad 12567, \quad 23457, \quad 13467, \quad 3567, \quad 12468, \quad 13458, \quad 12378.$$

Indeed, the determinant of the 8×8 matrix with rows $(1, x_1, \dots, x_7)$ for these supports is 8. Since $0, e_1, \dots, e_8$ are feasible, the knapsack polytope is full-dimensional in $\mathbb{R}^8$. Thus a face containing eight affinely independent points has dimension 7, so (4) defines a facet. $\square$

There are no within-class swaps in this example because all candidate coefficients are distinct. The proposed two-anchor rank test omits 12378, since every two-coordinate slice through it has a lower-cost witness. All twelve lower tight supports satisfy (3), whereas 12378 does not. Therefore the omitted point is needed for full rank, and the anchor algorithm rejects a true facet.

5 A correct rank formula

The false anchor theorem can be replaced by an exact rank computation using second moments. This does not by itself eliminate the counting problem, but it avoids all unsupported spanning claims.

Let

$$V = \{0\} \cup \{(j, r) : 1 \leq j \leq K, 1 \leq r \leq n_j\}$$

index the augmented coordinate 0 and all original variables. For $z \in \mathcal{Z}$, define

$$\rho(z) = (1, p(z)) \in \mathbb{R}^V.$$

For each feasible swap (j, r, s) , $r < s$, define

$$\sigma_{jrs} = (0, e_{jr} - e_{js}) \in \mathbb{R}^V.$$

Let

$$G = \sum_{z \in \mathcal{Z}} \rho(z) \rho(z)^\top + \sum_{(j,r,s) \text{ feasible}} \sigma_{jrs} \sigma_{jrs}^\top. \quad (5)$$

Theorem 5.1 (Gram rank criterion). *Assume the inequality is valid and supporting. Then $c^\top x \leq d$ defines a facet of the reduced knapsack polytope if and only if*

$$\text{rank}(G) = n.$$

Proof. Let A be the matrix whose rows are all vectors $\rho(z)$ for $z \in \mathcal{Z}$ and all feasible swap vectors σ_{jrs} . Then $G = A^\top A$. Over $\mathbb{R}$, $\ker(G) = \ker(A)$, because $v^\top G v = \|A v\|_2^2$. Hence $\text{rank}(G) = \text{rank}(A)$.

By Lemma 3.1, the row span of A is exactly $\text{span}\{(1, x) : x \in T\}$. Since the reduced polytope is full-dimensional and all tight points lie in one proper hyperplane, the inequality is facet-defining exactly when the augmented tight vectors have rank n . $\square$

It remains to compute the entries of G without enumerating $\mathcal{Z}$. For lower-bound thresholds $\ell = (\ell_1, \dots, \ell_K)$ with $0 \leq \ell_j \leq n_j$, define

$$N(\ell) = |\{z \in \mathcal{Z} : z_j \geq \ell_j \text{ for every } j\}|. \quad (6)$$

Only thresholds with at most two nonzero entries are needed. Specifically,

$$\begin{aligned} G_{00}^{\text{prof}} &= N(0, \dots, 0), \\ G_{0,(j,r)}^{\text{prof}} &= N(\ell^{j,r}), \\ G_{(i,r),(j,s)}^{\text{prof}} &= \begin{cases} N(\ell^{i,r} + \ell^{j,s}), & i \neq j, \\ N(\ell^{j, \max\{r,s\}}), & i = j, \end{cases} \end{aligned}$$

where $\ell^{j,r}$ has r in coordinate j and zero elsewhere. The swap contribution

$$\sum \sigma_{jrs} \sigma_{jrs}^\top$$

is computed directly from the feasible swap list in polynomial time.

6 Computing the threshold counts by epigraph finite differences

This section gives a precise counting reduction for $N(\ell)$.

Lemma 6.1 (Epigraph formulation). *For each j and each $q = 0, \dots, n_j - 1$, let*

$$L_{jq}(t) = F_j(q) + a_{j,q+1}(t - q).$$

For integral $t \in \{0, \dots, n_j\}$,

$$F_j(t) = \max_{0 \leq q < n_j} L_{jq}(t).$$

Proof. The slopes of the linear interpolation of F_j are the nondecreasing numbers $a_{j1}, \dots, a_{jn_j}$. Hence each line L_{jq} is a supporting line to this convex piecewise-linear interpolation, and the maximum of all such supporting lines equals the interpolation. Evaluating at integer t gives $F_j(t)$. $\square$

Fix a threshold vector ℓ . For an integer B , define the epigraph polytope

$$\begin{aligned} Q_\ell(B) = \{ (z, y) \in \mathbb{R}^{2K} : & \ell_j \leq z_j \leq n_j \quad (j = 1, \dots, K), \\ & \gamma^\top z = d, \\ & y_j \geq L_{jq}(z_j) \quad (j = 1, \dots, K, \ q = 0, \dots, n_j - 1), \\ & \sum_{j=1}^K y_j \leq B \}. \end{aligned} \tag{7}$$

Let

$$E_\ell(B) = |Q_\ell(B) \cap \mathbb{Z}^{2K}|,$$

with the convention $E_\ell(B) = 0$ when $B < 0$.

Lemma 6.2 (Finite-difference count). *For every threshold vector ℓ ,*

$$N(\ell) = \sum_{h=0}^K (-1)^h \binom{K}{h} E_\ell(b - h). \tag{8}$$

Proof. Fix an integer z satisfying the bounds $\ell \leq z \leq n$ and $\gamma^\top z = d$. By Lemma 6.1, the least integral y_j satisfying the j th epigraph inequalities is $F_j(z_j)$. Thus, if $F(z) \leq B$, the number of integer vectors y with $(z, y) \in Q_\ell(B)$ is the number of nonnegative integer slack vectors $s_j = y_j - F_j(z_j)$ satisfying $\sum_j s_j \leq B - F(z)$, namely

$$\binom{B - F(z) + K}{K}.$$

If $F(z) > B$, there are no such y . Therefore

$$E_\ell(B) = \sum_{\substack{\ell \leq z \leq n, \\ \gamma^\top z = d \\ F(z) \leq B}} \binom{B - F(z) + K}{K}.$$

The K th backward difference of the polynomial sequence $S \mapsto \binom{S+K}{K}$ is 1 for every $S \geq 0$, with the convention that the binomial coefficient is zero for $S < 0$. Applying the K th backward difference at $B = b$ gives exactly one contribution from every z with $F(z) \leq b$ and no contribution from the others, proving (8). $\square$

Thus each entry of the profile part of the Gram matrix is computable from $K + 1$ ordinary integer-point counts in rational polytopes of dimension $2K$ with $O(n + K)$ inequalities and coefficient bit-length polynomial in the input length.

Theorem 6.3 (Rank-counting reduction). *Let L be the binary encoding length of the knapsack-facet instance. There is an exact deterministic Turing reduction from the facet-recognition problem above to*

- *one fixed-dimensional separable-convex optimization call for validity;*
- *$O(n^2)$ fixed-dimensional separable-convex feasibility calls for feasible swaps;*
- *$O(Kn^2)$ ordinary integer-point counting calls for polytopes in dimension $2K$ with $O(n + K)$ inequalities and bit-length $\text{poly}(L + n)$;*
- *exact rational rank computation of an $(n + 1) \times (n + 1)$ integer matrix.*

Consequently, if the stated integer-point counting primitive in dimension q is solvable in time $f(q) \text{poly}(M, L')$ for polytopes with M inequalities and encoding length L' , then knapsack-facet recognition for inequalities with K distinct positive coefficients is fixed-parameter tractable in K .

Proof. Use Proposition 2.1 to reject invalid inequalities and nonsupporting inequalities. Test all feasible swaps as described after Lemma 3.1. For every threshold vector needed by Section 5, compute $N(\ell)$ by Lemma 6.2; there are $O(n^2)$ such thresholds, each using $K + 1$ counts. Assemble G from (5) and the formulas following (6). Theorem 5.1 gives the rank criterion. The stated running-time implication is immediate from the number and size of the oracle calls and from polynomial-time exact rank computation over $\mathbb{Q}$. $\square$

Remark 6.4 (Why this is not claimed as a full solution). Barvinok-type algorithms count integer points in rational polytopes in polynomial time when the dimension is fixed, and Barvinok–Woods extend the framework to projections [2]. Such fixed-dimension statements recover polynomial-time solvability for every fixed K . The parameterized question, however, asks for a uniform bound of the form $f(K) \text{poly}(n, L)$, with an exponent independent of K . The reduction above isolates the exact primitive needed for such a conclusion, but the present note does not prove the required uniform counting bound.

7 Certificates

The Gram criterion is primarily an algorithmic rank test, but it also preserves ordinary certificates.

For a yes-instance, a certificate consists of n tight binary points whose augmented matrix has rank n . Such a certificate can be recovered from a rank basis of the matrix whose Gram matrix is G if the counting oracle also supplies enough witnesses, or by self-reduction using rank tests with rows constrained to threshold regions. For practical verification, the certificate is simpler: list the n binary tight points and one nonzero $n \times n$ determinant. Each listed point is checked directly against $a^\top x \leq b$ and $c^\top x = d$, and the determinant is checked by exact arithmetic.

For a no-instance returned by the rank test, one may certify non-facetness by giving a nonzero vector in the nullspace of G . This vector defines a linear equation satisfied by every canonical tight profile and every feasible swap direction; through Lemma 3.1, it is satisfied by every tight point, proving that the tight face has dimension at most $n - 2$.

8 Conclusion

The proposed two-anchor proof cannot be repaired locally: its central spanning theorem is false, and the associated algorithm fails on an explicit valid facet. A correct replacement is to compute

the tight affine rank through a Gram matrix. The required matrix entries are threshold counts of the convex count-profile set; these counts reduce by a finite-difference identity to ordinary lattice-point counts in $2K$ -dimensional epigraph polytopes.

Thus the present state is partial progress. The open FPT question is reduced to a concrete fixed-dimensional counting primitive tailored to separable-convex epigraphs. Proving a uniform $f(K)$ $\text{poly}(n, L)$ bound for that primitive, or finding a way to avoid exact counting altogether, would complete the solution.

References

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